

Retinal infection and immune recovery

Part 2 | Compare necrotising retinitis patterns

Appearance and immune context narrow the investigation.

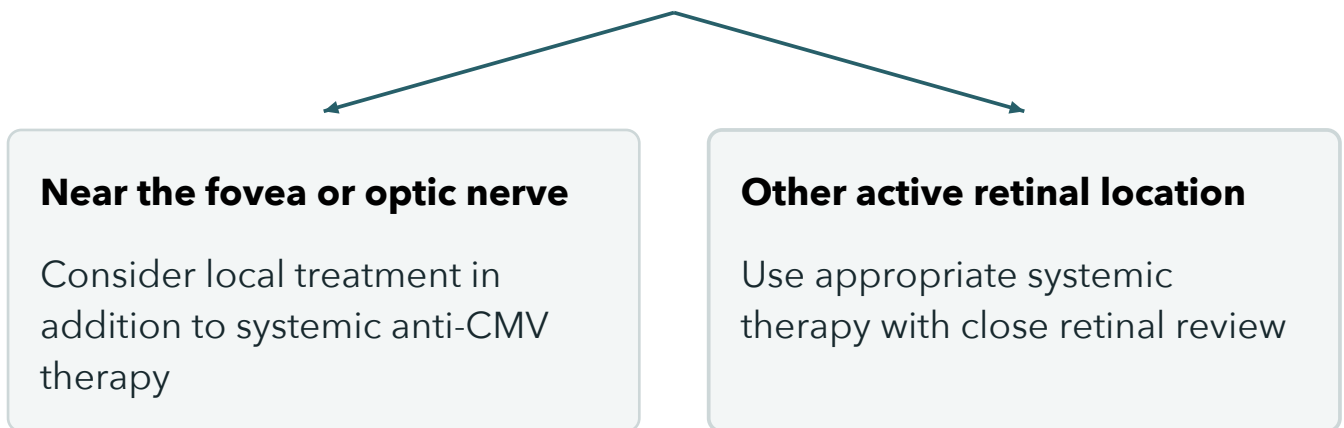
Finding	Implication
CMV in severe immunosuppression	Necrotising haemorrhagic retinitis; often limited vitritis
ARN	Peripheral necrosis, arterial occlusion and prominent inflammation
Toxoplasmosis	Focal active lesion, often beside a pigmented scar
Atypical or treated eye	Consider ocular fluid testing and alternative causes

No single descriptive sign is fully specific. Map the lesion, degree of vitritis, immune status and tempo, then identify an investigation that can change the diagnosis or drug choice. Prior treatment can alter the phenotype. A negative blood result is not a universal exclusion of local infection, especially in an immunocompromised patient.

Active CMV: local control and systemic protection

Sight-threatening location changes urgency and delivery.

Is active disease immediately threatening central vision?



Systemic treatment also protects the fellow eye and extraocular sites. Local treatment alone does not provide that protection. Coordinate immune restoration and infectious-disease care while documenting lesion activity and functional risk. The chosen route depends on location, severity, tolerance and ability to attend, rather than a blanket local-injection rule.

Immune recovery uveitis versus reactivation

A previously treated lesion can coexist with new inflammatory blur.

Finding	Implication
New advancing necrotic border	Reconsider active or recurrent infection
Inactive scar after immune restoration	Inflammatory uveitis or macular oedema may dominate
OCT change	Locate oedema and other macular causes
Treatment choice	Control inflammation while maintaining appropriate antiviral coverage

Do not equate new vitritis after ART with antiviral failure. Conversely, do not give unopposed immunosuppression without checking lesion activity. At IRU onset the NIH guidance favours both anti-CMV treatment and corticosteroids; later inflammatory oedema needs a tailored approach. Document the retinal border and immune trajectory before selecting therapy.

Drug selection includes toxicity monitoring

An ocular response does not establish systemic treatment safety.

Finding	Implication
Valganciclovir or ganciclovir	Blood-count and renal considerations
Foscarnet	Renal and electrolyte monitoring
Intravitreal cidofovir	Avoid; current NIH guidance contraindicates it
Prolonged treatment	Review interactions and adherence with the medical team

Older teaching material can contain unsafe obsolete local regimens. Use current prescribing information and specialist protocols, not a remembered historical dose. Explain how baseline organ function, changing blood results and concurrent medicines affect the plan. A new systemic abnormality may require coordinated modification even when the retinal lesion is improving.

After infection: watch for structural sequelae

Inactive retinitis does not end retinal follow-up.

Finding	Implication
Necrotic peripheral retina	Breaks and late retinal detachment
ARN after acute control	Continue surveillance of both eyes
Scarring near fixation	Functional limitation despite inactive disease
New symptoms	Reassess activity and mechanical complications

Explain the difference between controlling an organism and restoring damaged retina. Surveillance should look for recurrent activity, traction, breaks, detachment and macular consequences. Preventive procedures are not automatically indicated in every ARN eye; describe the actual anatomy and evidence limitations. The follow-up plan should make new floaters, field loss or blur actionable.